



FACULTY OF COMPUTING

SEMESTER II - 2024/2025

SECV4213-02 PEMBANGUNAN PERMAINAN KOMPUTER (COMPUTER GAMES DEVELOPMENT)

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Game Idea and Gamification Plan

The game Forest Cleanup turns environmental consciousness into a first-person, interactive experience. Utilizing WASD controls for walking and right-click for sprinting, players traverse a 100x100 forest area in search of apples (red spheres) and plastics (gray cubes). For the sake of efficient gameplay, each session is timed with a 60-second countdown.

Key gamification elements:

- **Scoring System:** Players earn points by picking up plastics (good) and lose if they collect all apples first.
- **Race Condition:** Collect all plastics before apples for a win; the reverse results in a loss.
- **Countdown Timer:** A visible 60-second timer drives urgency.
- **Visual & Audio Feedback:** Collecting apples triggers a white-smoke effect and a bespoke apple sound; collecting plastics triggers brown-smoke; win, lose, and background music reinforce milestones.

Technical Implementation Overview

The entire game is set in Babylon.js as a textured ground plus skybox creates the scene, invisible boundary walls enclose the play area, and a UniversalCamera with gravity and collision makes movement feel realistic. Particle systems manage smoke effects, and collectibles are basic mesh primitives that are identified as pickable. AdvancedDynamicTexture is used to draw the on-screen score and timer, while hardware scaling and frozen static models are used to optimize performance for fluid frame rates.

Reflections and Challenges Faced

The main difficulties were adjusting collision distances for dependable pickups, combining visual and aural richness with browser performance, and making sure particle systems and meshes are disposed of appropriately to avoid memory leaks. It was also crucial to refine UI positioning for device clarity in order to provide a polished, captivating experience.

Video Demo Link: <https://youtu.be/jRqdAGumx9U?si=fWf9iowpqDPPuRZE>

GitHub Link: <https://github.com/Ridzuanbkr/GDTask6materials.git>